Enhancing Data and Algorithm in Offline Reinforcement Learning

Project Report of the course ‘Deep Reinforcement Learning’

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- Our Idea
- Preliminary Experiments
- Improvements
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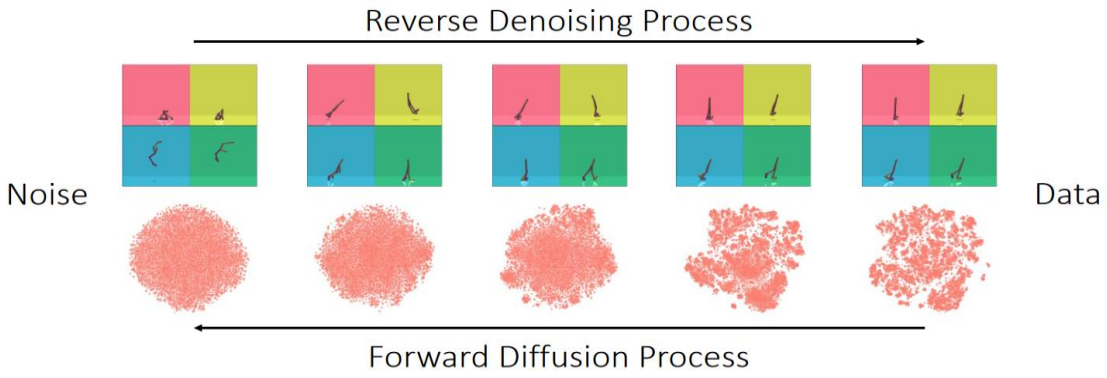
Literature Review

- Offline RL algorithm
  - BCQ
  - CQL
  - IQL
  - SAC-N
- Dataset trimming strategy
  - DiffStitch
  - Exploration by RND
  - Survival Instinct
- Synthetic trajectories with Diffusion models
  - Synthetic Experience Replay (SER)
  - Generative Trajectory Augmentation (GTA)
  - Policy-Guided Diffusion (PGD)

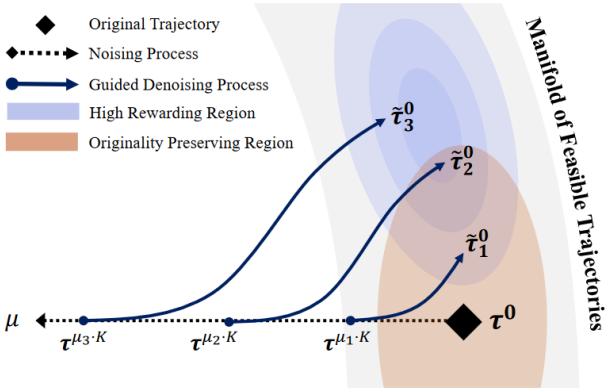
Literature Review—Diffusion

- SER
  - **Unconditionally** generates synthetic experiences and upsample data for training, at **transition** level.
  - Verifies that **periodically** updating dataset is better than **continuously** updating dataset, but not compared with updating once and for all.
  - Verifies that when original dataset is good, the **mixture** of real and synthetic dataset is **not better** than completely synthetic dataset.
  - Improves the performance on small datasets and enhances data efficiency in online settings.



Literature Review—Diffusion

- GTA
  - Partially noises **original** trajectories, and denoises **conditioning** on **amplified return** value, at **trajectory** level.
  - Trains RL on the dataset generated **once and for all**.
  - Verifies that **trajectory** level is **better** than transition level.
  - Enhances the performance of widely used offline RL algorithms in both dense and sparse reward settings, and improves the quality of the data.

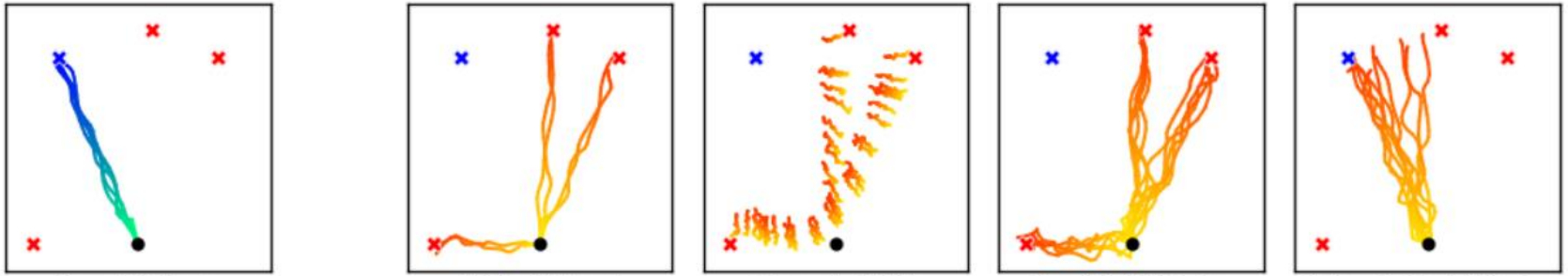


$$\hat{y}(\boldsymbol{\tau}^0, \alpha) = \alpha \cdot \sum_{i=t}^{T} \gamma^{i-t} r_i,$$

- Noising controls level of **exploration**.
- Amplified parameter controls **conversion** towards a high-rewarding trajectory region.
- Use reward of original trajectories to prevent **extreme value**.

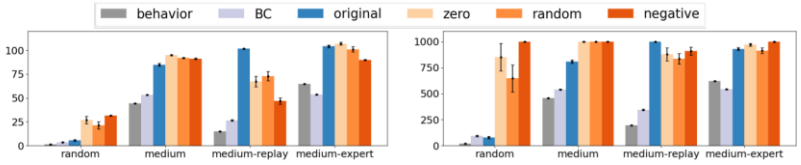
Literature Review—Diffusion

- PGD
  - Applies **guidance from the target policy** to generate synthetic **trajectories** that are
    (a.) more representative of the **target** policy, while
    (b.) do not diverge too far from the **behavior** policy.
  - **Periodically** generates trajectories and **finetune** offline RL algorithms trained from original datasets.
  - Addresses the issue of **distribution shift** between the target and behavior policies



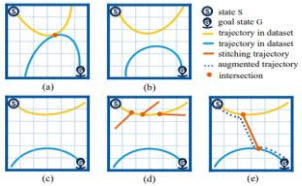
Our Idea

- (Survival Instinct, 2023-11) tells us **offline RL is not sensitive** to the **manipulation of reward** signal.

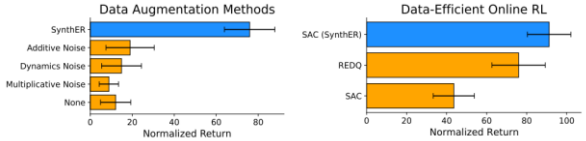


(a) Normalized scores (b) Episode lengths

- **Diffusion model** can be used to generate **high-quality trajectories**.(DiffStitch,2024-2)



- **Diversity** of diffusion models can enhance the **generalization** of offline RL with the blessing of function approximation error (SER, 2023-10)



(a) IQL [41] on a reduced 15% subset of walker2d medium-replay [21]. (b) SAC [24] on 6 DeepMind Control Suite [71] and OpenAI Gym [10] environments.

Our Idea: diversity, fidelity, high reward

- We build upon GTA(Generative Trajectory Augmentation) to **augment the dataset** with **high return trajectories**.

- However, using synthetic data **harms fidelity**, compared with solely relying on original data.

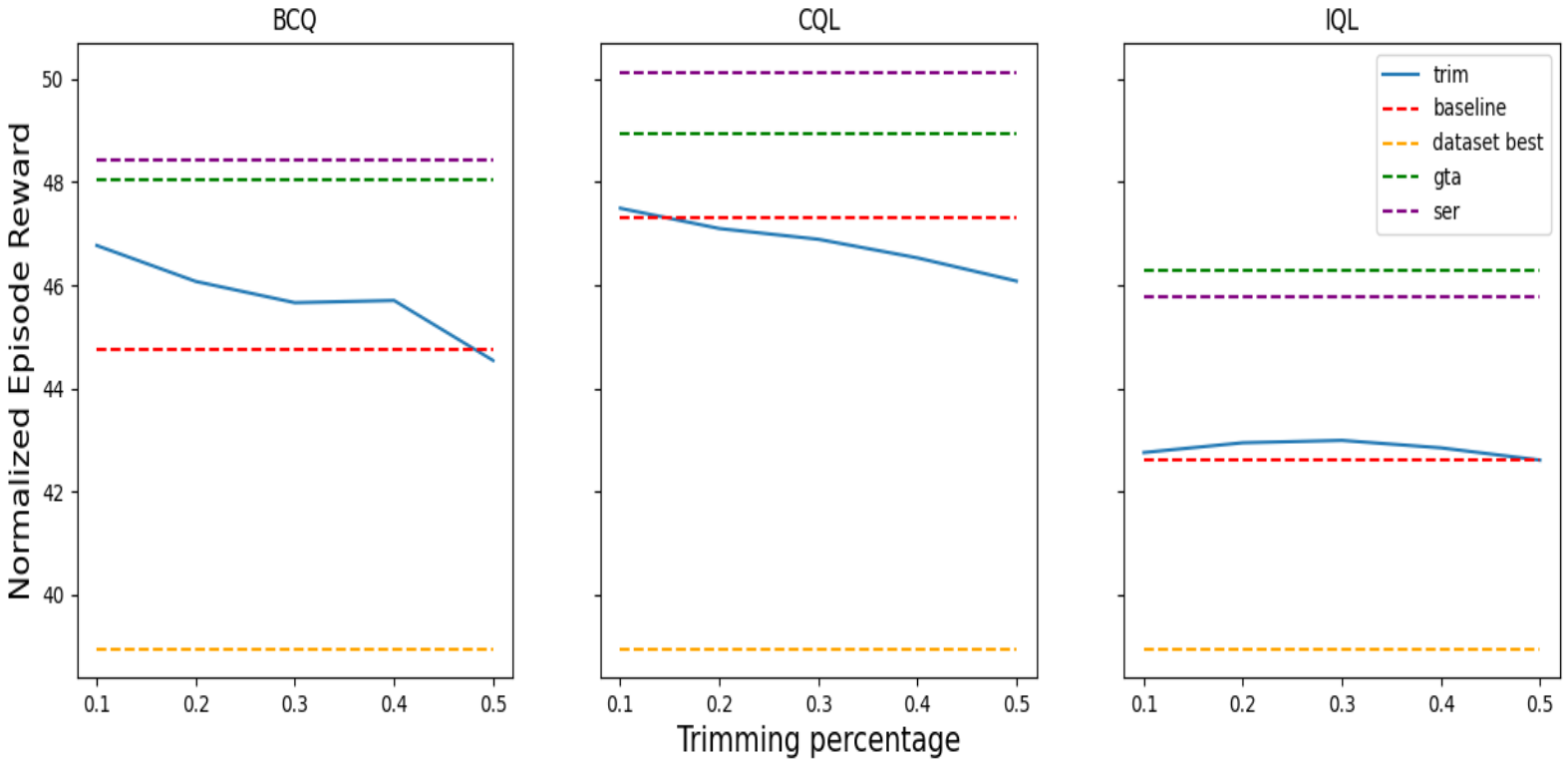
- **Our approach**: use the **mixture of synthetic and original dataset**
  Different from GTA.

- To **further enhance the fidelity of the synthetic data**, we suggest to do the following
  - train a **forward dynamics model** to filter and trim the generated dataset.
  - Train an **ensemble diffusion models** to get more accurate trajectories, and use the model uncertainty to perform importance sampling.

Design

- Baseline
  - Run **CQL, BCQ, IQL** with 0%, 10%, 20%, 30%, 40% and 50% trim percentages respectively.
  - Run **EDAC** with 0%, 10%, 20%, 30%, 40%, 50%, 60%, 80%, 100% trim percentages.
- Advanced
  - Generate high-return pseudo trajectories with **conditional diffusion** model, and rerun baseline with the generated dataset with no trim. (Following GTA)
  - Learn the policy with the **mixture of the generated dataset and the original dataset**. (Running currently).

Results—d3rlpy Algorithms: BCQ, CQL, IQL



Results—EDAC

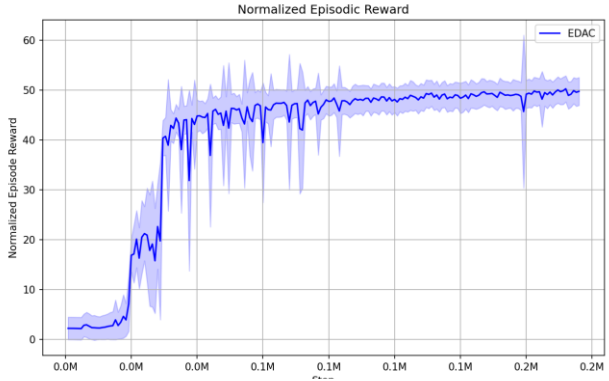
We choose an **advanced offline RL algorithm**: **EDAC**, An **efficient version of SAC-n**.

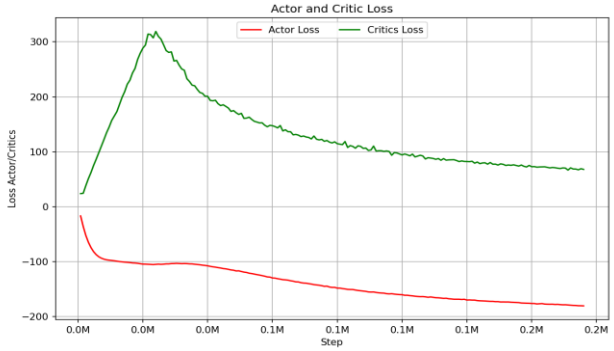
(a) N critic nets (over-estimation)

(b) Maximize ensemble diversity (minimized cosine similarity)

$$\underset{\phi}{\text{minimize}}\ J_{\text{ES}}(Q_\phi) := \mathbb{E}_{\mathbf{s},\mathbf{a}\sim\mathcal{D}}\left[\frac{1}{N-1}\sum_{1\le i\ne j\le N}\underbrace{\langle \nabla_{\mathbf{a}} Q_{\phi_i}(\mathbf{s},\mathbf{a}), \nabla_{\mathbf{a}} Q_{\phi_j}(\mathbf{s},\mathbf{a})\rangle}_{\text{ES}_{\phi_i,\phi_j}(\mathbf{s},\mathbf{a})}\right].$$

Concretely, our final objective can be interpreted as measuring the pairwise alignment of the gradients using cosine similarity, which we denote as the Ensemble Similarity (ES) metric $\text{ES}_{\phi_i,\phi_i}(\mathbf{s},\mathbf{a})$, and





Algorithm 1 Ensemble-Diversified Actor Critic (EDAC)

1: Initialize policy parameters θ, Q-function parameters $\{\phi_j\}_{j=1}^N$, target Q-function parameters $\{\phi'_j\}_{j=1}^N$, and offline data replay buffer $\mathcal{D}$

2: **repeat**

3: Sample a mini-batch $B = \{(\mathbf{s},\mathbf{a},r,\mathbf{s}')\}$ from $\mathcal{D}$

4: Compute target Q-values (shared by all Q-functions):

$$y(r,\mathbf{s}') = r + \gamma\left(\min_{j=1,\dots,N} Q_{\phi'_j}(\mathbf{s}',\mathbf{a}') - \beta\log\pi_\theta(\mathbf{a}'\mid\mathbf{s}')\right), \quad \mathbf{a}'\sim\pi_\theta(\cdot\mid\mathbf{s}')$$

5: Update each Q-function Q_{ϕ_i} with gradient descent using

$$\nabla_{\phi_i}\frac{1}{|B|}\sum_{(\mathbf{s},\mathbf{a},r,\mathbf{s}')\in B}\left(\left(Q_{\phi_i}(\mathbf{s},\mathbf{a}) - y(r,\mathbf{s}')\right)^2 + \frac{\eta}{N-1}\sum_{1\le i\ne j\le N}\text{ES}_{\phi_i,\phi_j}(\mathbf{s},\mathbf{a})\right)$$

6: Update policy with gradient ascent using

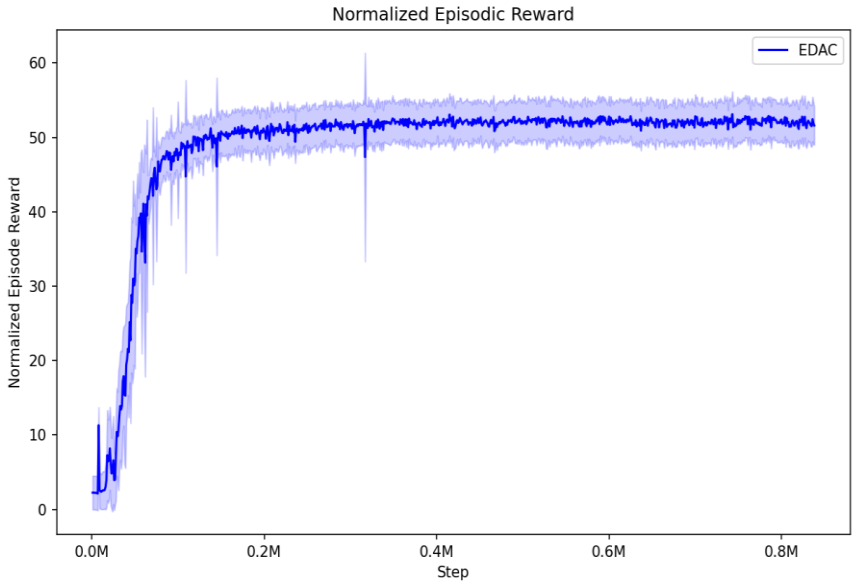
$$\nabla_\theta\frac{1}{|B|}\sum_{\mathbf{s}\in B}\left(\min_{j=1,\dots,N} Q_{\phi_j}(\mathbf{s},\tilde{\mathbf{a}}_\theta(\mathbf{s})) - \beta\log\pi_\theta(\tilde{\mathbf{a}}_\theta(\mathbf{s})\mid\mathbf{s})\right),$$

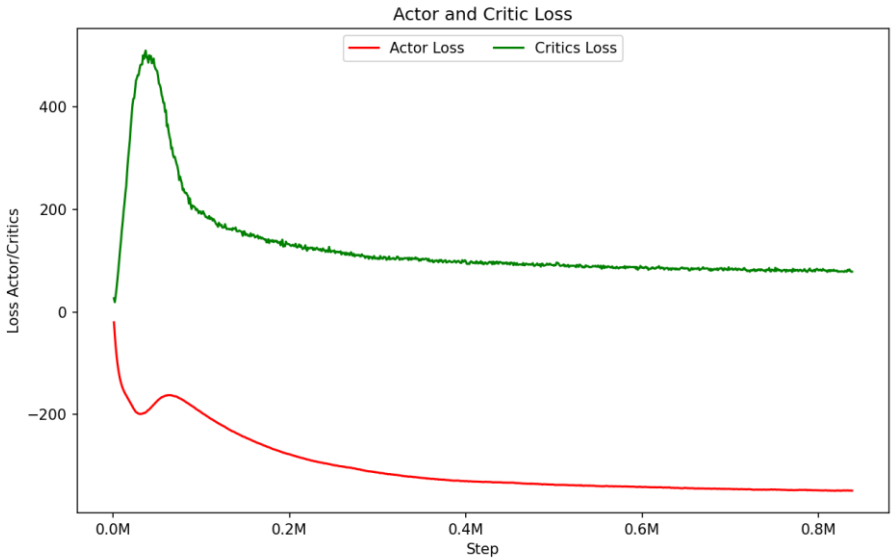
where $\tilde{\mathbf{a}}_\theta(\mathbf{s})$ is a sample from $\pi_\theta(\cdot\mid\mathbf{s})$ which is differentiable w.r.t. θ via the reparametrization trick.

7: Update target networks with $\phi'_i \leftarrow \rho\phi'_i + (1-\rho)\phi_i$

Advanced Results—EDAC

Training an EDAC agent with the **untrimmed** dataset completely **generated by diffusion model (5 M short trajectories)**
Algorithm converges fast and yields normalized reward **more than 50**
Surpassing all three classical algorithms trained on the same synthetic data.

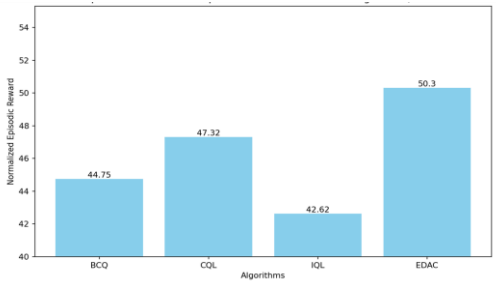




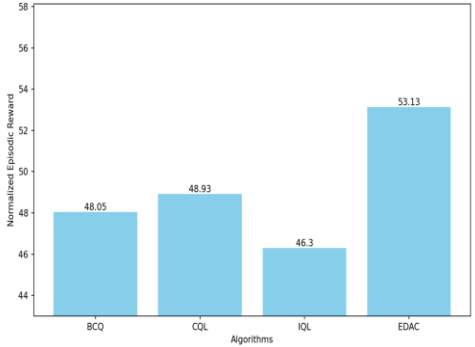
The effect of synthetic dataset on offline RL algorithms

EDAC wins by a large margin on **large synthetic dataset.**
BCQ and CQL share similar spirit, while IQL is lower by a large margin.

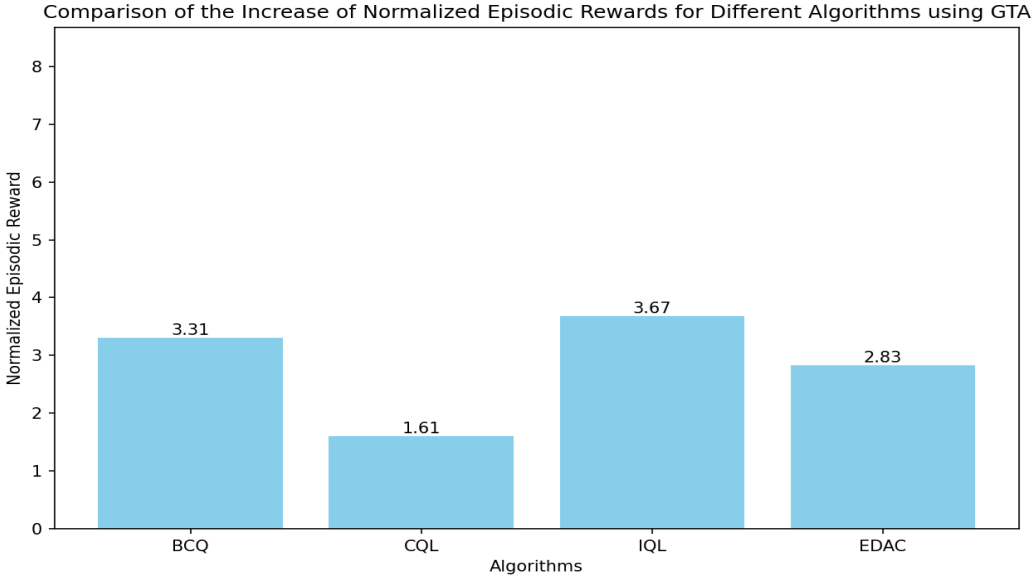
on the original dataset



on the synthetic dataset

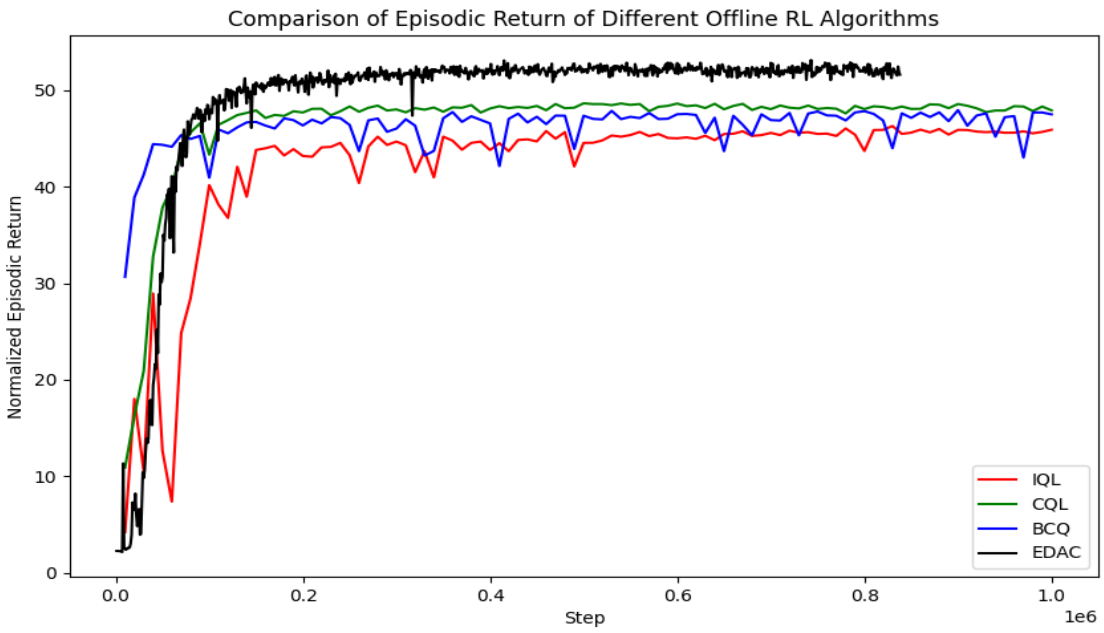


Synthetic data is not the most beneficial configuration for EDAC algorithm. **IQL is the one benefits the most from augmentation.**

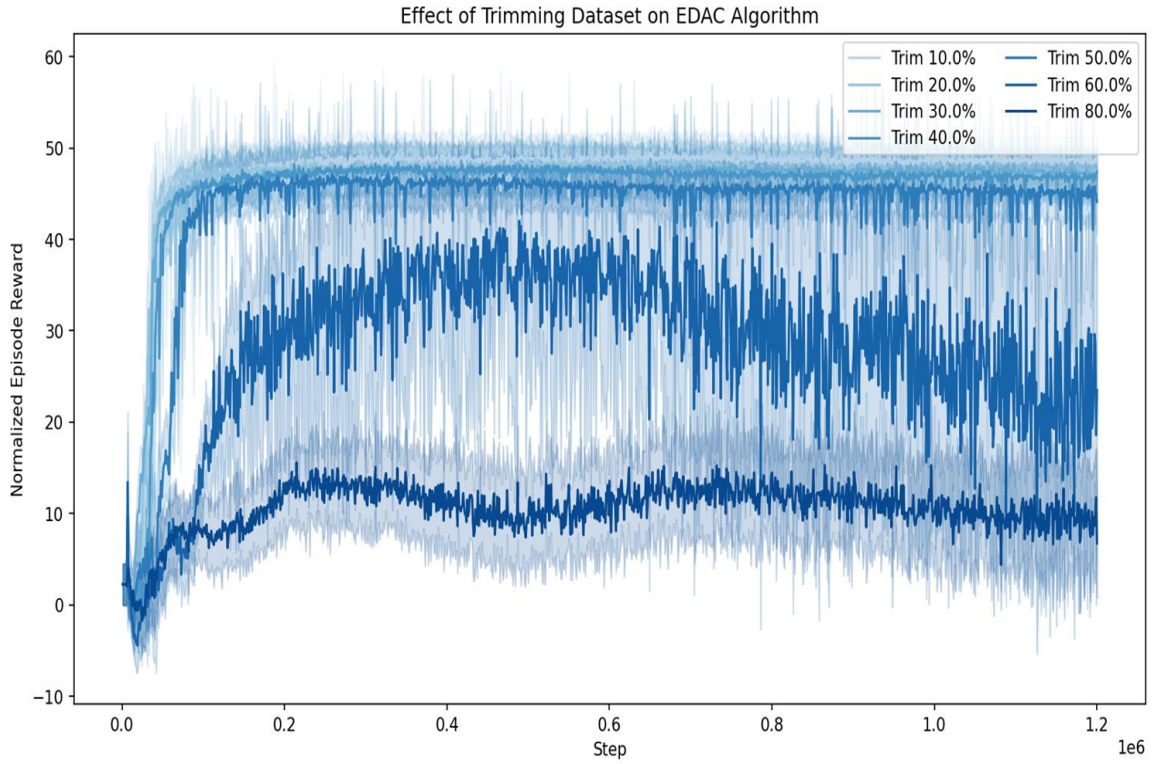


The effect of synthetic dataset on offline RL algorithms

EDAC converges faster on the **large amount of synthetic data** than the other three algorithms, with IQL being the slowest to converge.



Trimming original dataset deteriorates EDAC's performance



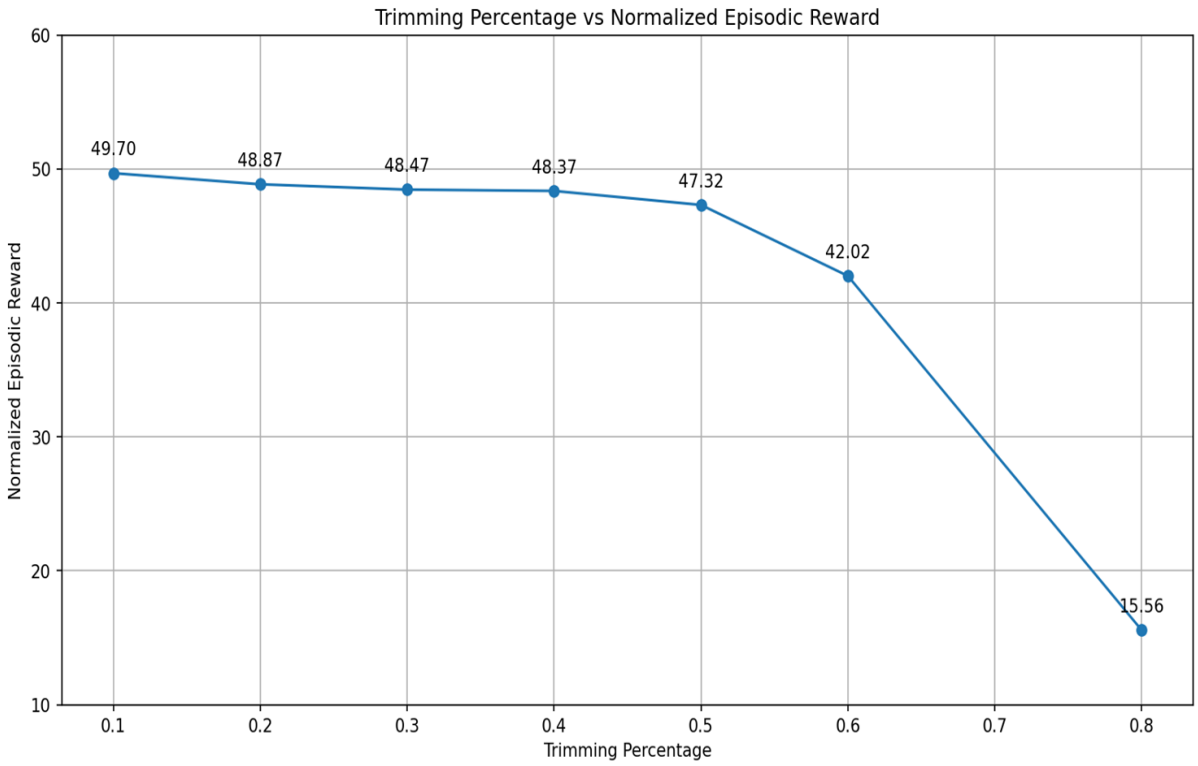
Trimming the original dataset According to rewards.

Method:
Drop the lowest rewarding episodes by percentage.
Could cause the train set to shrink.

Discovery:
Trimming too hard makes agent learn suboptimal policies.

Analysis:
Lack of sufficient environmental information from sparse samples.

Trimming original dataset deteriorates EDAC's performance

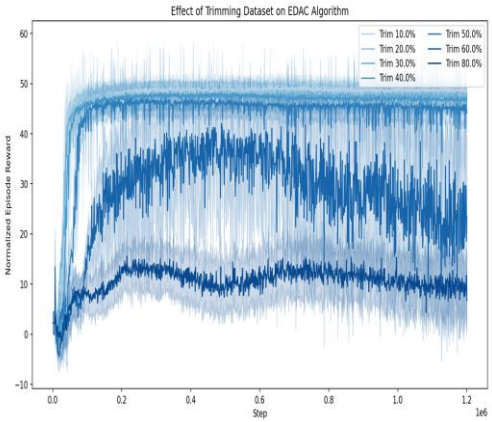


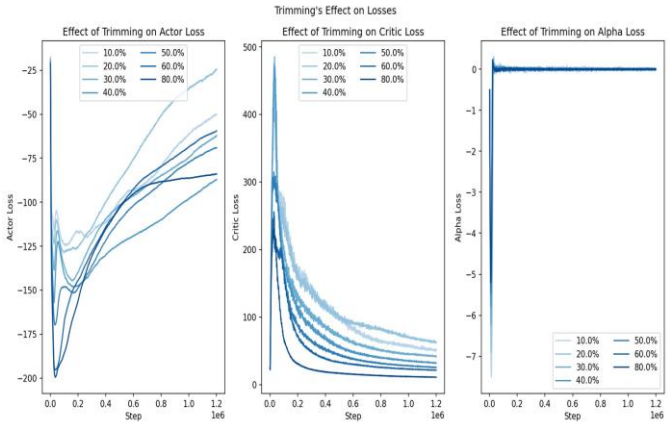
Trimming the original dataset According to rewards.

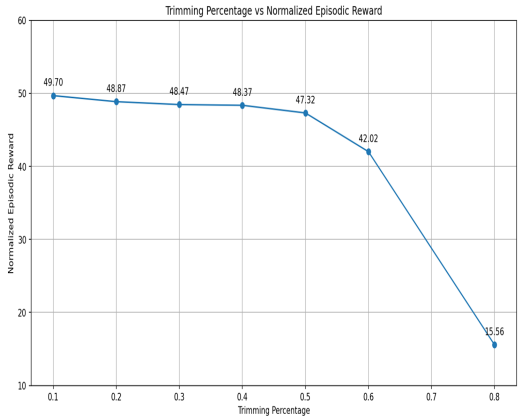
Discovery:
Trimming with a small percentage
(<40%) will not drastically decrease the performance.
Smooth curve in the beginning.

Application:
Possibly a way to **compress the storage of offline dataset without hurting performance too much.**

Trimming original dataset deteriorates EDAC's performance





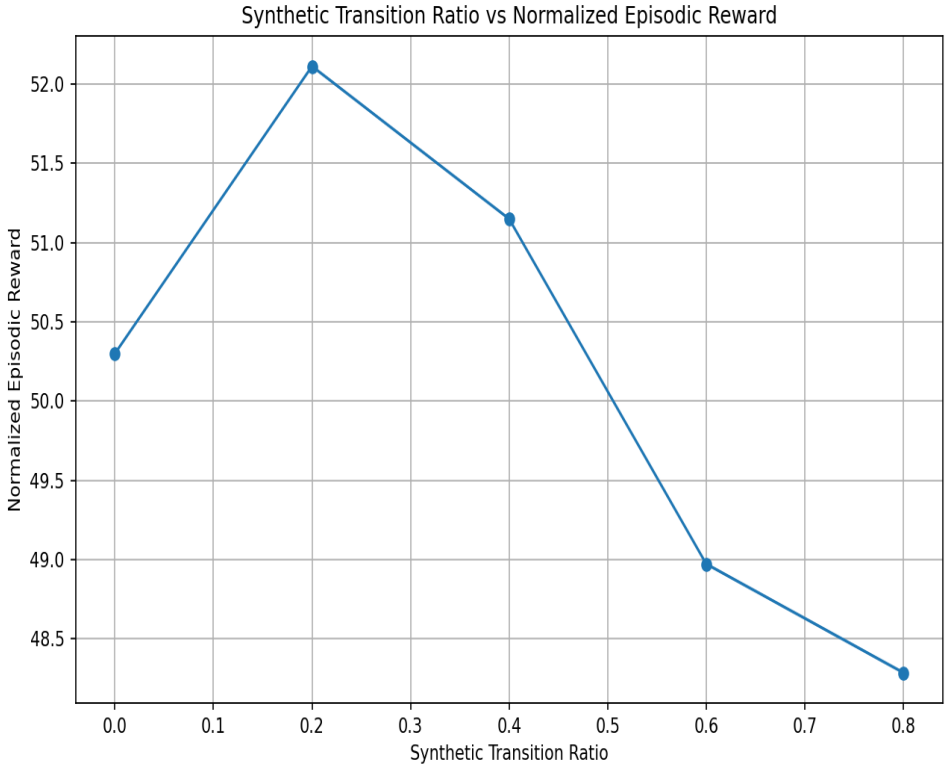


EDAC algorithm is highly **data-driven** in current setup.

Having 10 Q network heads, it possesses large amount of parameters makes it **data-hungry.**

Small dataset makes it **overfit** quickly.

Merging original dataset with synthetic dataset for EDAC



Preliminary Experiment failed…

Scatter all the episodes from both datasets

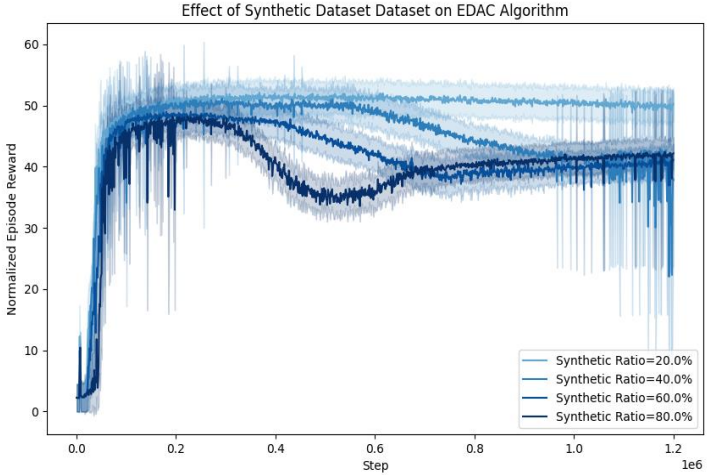
Directly merge the transitions without shuffle.

First train on fake data, then directly transfer to real data.

Result: better not to merge?

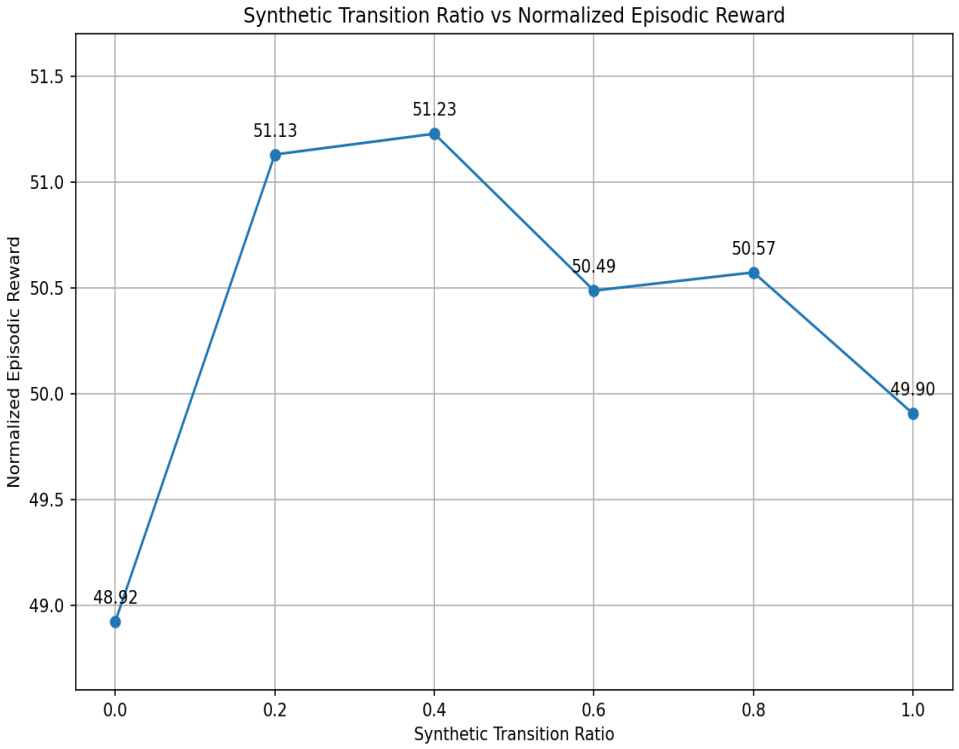
Rethink: GTA. Initial dist != stationary dist ?

Abrupt distribution shift hurts training ?



Literature → Our Idea → Design

Merging original dataset with synthetic dataset for EDAC

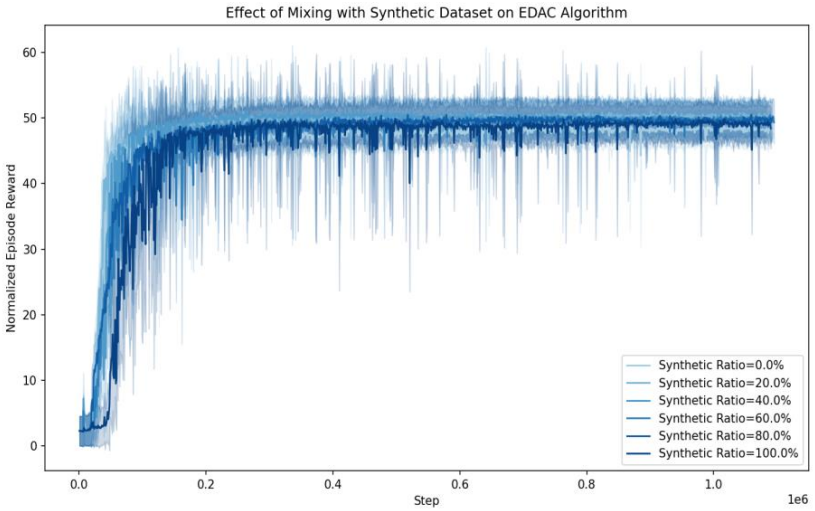


Mix original data with 0.0%, 20%, 40%, 60%, 80%, 100% of generated short trajectories.

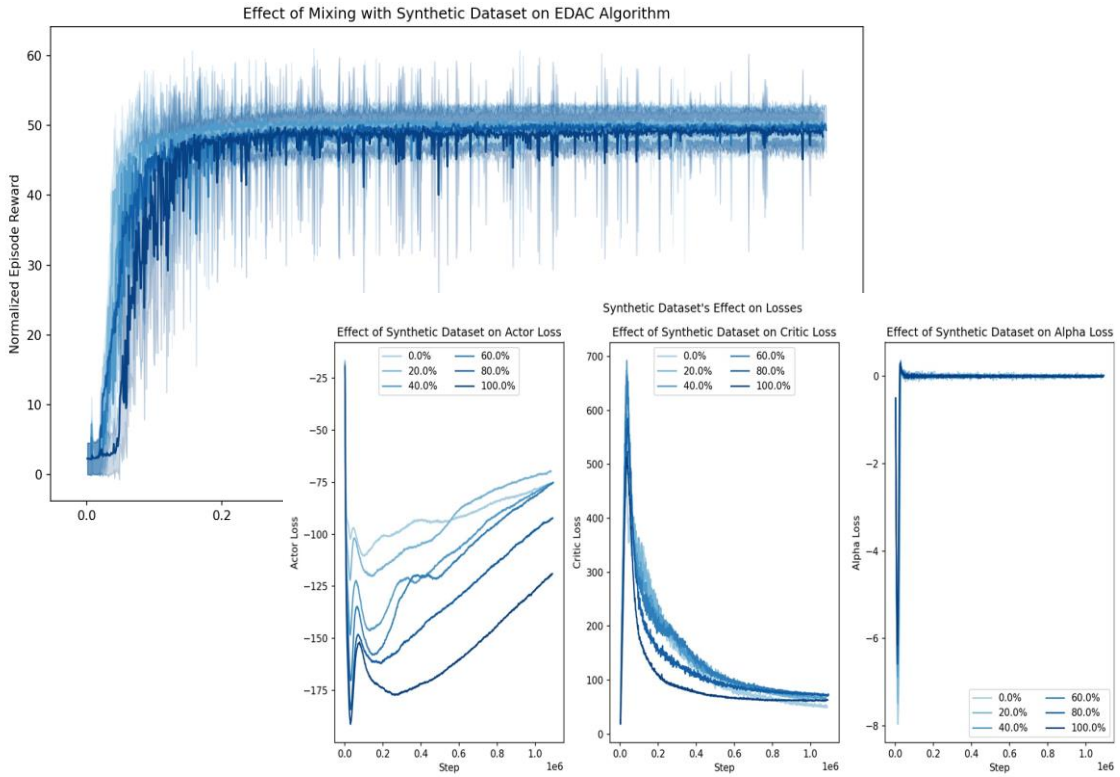
Literature → Our Idea → Design

Do the experiment again by…

(a) **Fix** the **total number of transitions**.
(b) First pick entire **episodes** from **diffusion** dataset. Because they are generated in the same batch.
(c) Then Split diffusion episodes to transitions.
(d) **Random shuffle** the short (30, fake) and long (1000, real) transitions **for a balanced mixture.**



Merging original dataset with synthetic dataset for EDAC



Do the experiment again by…
We find **all the groups converge normally after changing the mixture methodology.**

Conclusions:

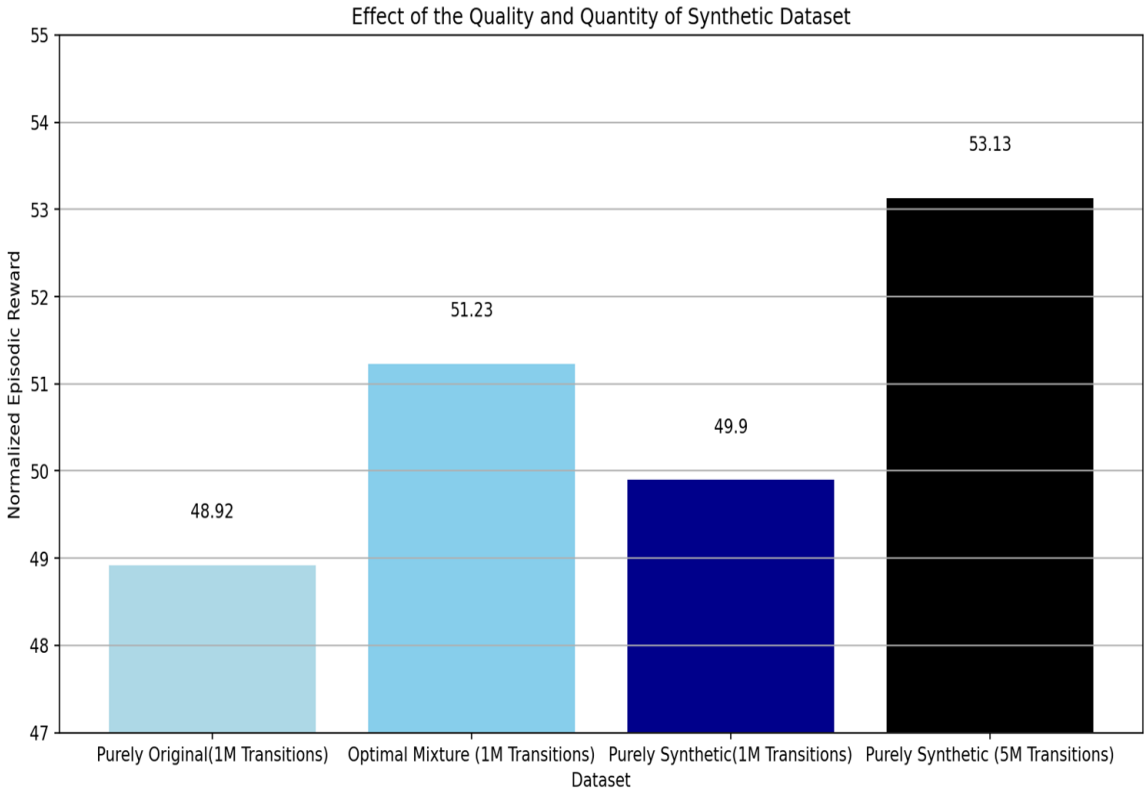
(a) Diversity of data matters in Offline RL, not just quantity.

(b) Mixture methodology matters, as transfer learning needs to avoid abrupt distribution shift.

All fake -> All real
Fake, Real, Fake, Real, …

Since we reduced number of transitions than previous experiments, the optimal performance decreases by a small amount.

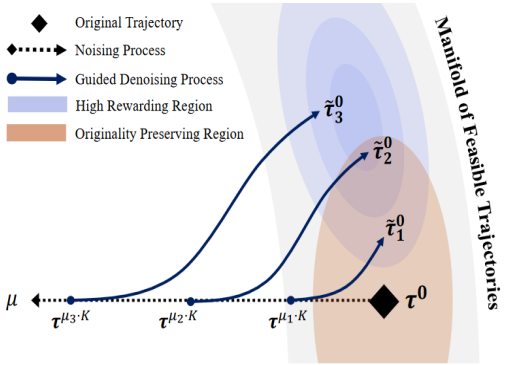
Merging original dataset with synthetic dataset for EDAC



Conclusions:

(a) Diversity(quality) of data matters in Offline RL, not just **quantity.**

(b) Mixture methodology matters, as **transfer learning** needs to **avoid abrupt distribution shift in datasets.**



Future Directions

- Improvement on fidelity
  - Train a **forward dynamics model**, and filter the pseudo trajectories. (Limitation mentioned in GTA)
  - **Ensemble** diffusion models and perform **importance sampling** for pseudo trajectories with weights of uncertainty.
- Improvement on GTA
  - Apply **periodically** updating to GTA
  - Combine **policy guidance** from PGD with GTA
  - See how diffusion models can help **online RL**, like RLPD.
- Evaluation
  - See how the **quality of original dataset** effect the mixture ratio choices. (SER claims with expert datasets, we should only use synthetic dataset)
  - Figure out the **exploration** on the offline dataset, with the process of **noising** in GTA.

Thanks for listening!

References

[1] Guanghe Li, Yixiang Shan, Zhengbang Zhu, Ting Long, and Weinan Zhang. Diffstitch: Boosting offline reinforcement learning with diffusion-based trajectory stitching, 2024.
[2] Cong Lu, Philip J. Ball, Yee Whye Teh, and Jack Parker-Holder. Synthetic experience replay, 2023.
[3] Junjie Zhang, Jiafei Lyu, Xiaoteng Ma, Jiangpeng Yan, Jun Yang, Le Wan, and Xiu Li. Uncertainty-driven trajectory truncation for data augmentation in offline reinforcement learning, 2023.
[4] Lee, Jaewoo, *et al*. "GTA: Generative Trajectory Augmentation with Guidance for Offline Reinforcement Learning." arXiv preprint arXiv:2405.16907 (2024).
[5] Jackson, M. T. *et al.* "Policy-Guided Diffusion." arXiv preprint arXiv:2404.06356 (2024).
[6] Philip J. Ball and Laura Smith and Ilya Kostrikov and Sergey Levine. "Efficient Online Reinforcement Learning with Offline Data." arXiv preprint arXiv:2302.02948 (2023).